# Jack Le (Tiep Le)

## CTO · Solutions & Technical Architect

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Vietnam · English, Vietnamese

### I architect platforms that stay correct under load — and lead the teams that ship them.

18 years from multi-threaded telecom systems to AI-native products. I sit between the business and the engineers: run the solution workshop, draw the C4, write the ADRs, then build and load-test the first version myself. From a 300-engineer delivery division to the CTO seat, across telecom, insurance, banking, healthcare, enterprise SaaS, gaming, Web3 and AgriTech. Today I design AI-native systems: agent memory over MCP, server-authoritative economies, deterministic real-time simulation.

| 300 | 96.0% | 10,500+ | 4,000 | 4 |
|---|---|---|---|---|
| engineer division I was Technical Architect for | recall@5 on LongMemEval-S, 500 questions | players in the first month after launch | CCU on 4 shards — one 4-vCPU box that also runs Postgres and Redis | countries on-site with enterprise clients |

## EXPERIENCE

**Principal Engineer & Architect** · Independent · AI & platform engineering — 2023 – now

Design and build production platforms end to end, from the first stakeholder conversation to the on-call runbook.

- Built Xóm Nông Trại end to end — game design, 3D, audio, code, release and operations — now live: ledger-backed economy, idempotent mutations, outbox, AI ops agents limited to proposals that humans approve.
- Created Cortex Hub, an open-source MCP platform for agent memory and code intelligence (96.0% recall@5 on LongMemEval-S).
- Built a deterministic Rust/WASM simulation and authoritative multiplayer server, benchmarked to 1.6 ms tick p50 at 64 players.
- Delivered a multi-tenant AgriTech platform and the Kèo Thơm real-time auction marketplace.

TypeScript · Node.js · Rust · PostgreSQL · Redis · GCP · Cloudflare · MCP · LLM

**Chief Technology Officer** · High-tech agriculture company — 2022 – 2023

Led architecture for a platform running thousands of hectares of production.

- Big-data pipelines processing millions of field data points for real-time monitoring and analytics.
- Crop lifecycle, export-compliance traceability and live operational dashboards on distributed services.
- Kept the system scalable and reliable under continuous data growth.

Microservices · Kubernetes · PostgreSQL · Big data · Real-time

**Chief Technology Officer** · D2 Solutions — 2021 – 2022

Founded and led the team behind a cross-platform 3D multiplayer game with a Web3 economy.

- Distributed real-time backend built to scale to millions of users with elastic micro-clusters of ~20 players per room.
- Custom low-latency TCP networking layer that removed the dependency on Photon.
- gRPC, Kafka and Redis on Kubernetes (HPA/VPA) with CI/CD and a full observability stack.
- Auditable on-chain / off-chain token economy.

C++ · Unity · gRPC · Kafka · Redis · Kubernetes · AWS · GCP

**Technical Architect** · Enterprise platforms — 2018 – 2021

Architecture for ERP, logistics and e-commerce platforms serving 100K+ users.

- Microservices on Kubernetes with PostgreSQL and CI/CD on AWS.

Microservices · Kubernetes · PostgreSQL · AWS · CI/CD

**Senior Software Engineer** · Ansarada — 2016 – 2018

Enterprise SaaS for deal management and virtual data rooms.

- Engineered inside the most disciplined microservices architecture I have worked in: small services, explicit contracts, CI/CD on every change.
- Test-driven development as the norm; kept unit-test coverage above 90% on the code I owned.

- Tuned SQL Server queries and indexes on performance-critical paths — work singled out for praise by the CTO.
- Moved from outsourcing to product engineering: features measured by customer value.

.NET · Microservices · SQL Server · TDD · CI/CD

**Technical Architect · Senior Engineer** · TMA Solutions 2008 – 2016

Grew from engineer to Technical Architect for the DG1 division (~300 engineers), serving global telecom clients.

- Owned architecture, technical direction and mentoring across the division.
- High-performance multi-threaded C/C++ on Windows, Linux and Solaris; led Windows → Linux and 32 → 64-bit ports down to drivers.
- On-site with customers in the USA (twice, six months each), the UK and Australia (twice).
- Outsourced delivery for banking and healthcare clients (client names under NDA).

C/C++ · Telecom · Linux · Solaris · Multi-threading

## SELECTED PROJECTS

**Xóm Nông Trại** · Founder · built end to end 2026

Server-authoritative social platform with a real economy

Built end to end: Gameplay · 3D & art · Music & sound · Build · Process · Operations

**10,500+** players in the first month after launch (Oct 2026)

**23 ms** API p95 in production at ~4.1M requests a day

TypeScript · Fastify · PostgreSQL 16 · Redis · BullMQ · Three.js — Play it live · nongtrai.jackle.dev

**Cortex Hub** · Creator 2026

One MCP endpoint for AI agent memory and code intelligence

**96.0%** recall@5 · LongMemEval-S (500 q)

**97.8%** recall@10 on the same set

TypeScript · MCP · Qdrant · SQLite · Ollama — github.com/lktiep/cortex-hub

**RA2 Studio** · Sole engineer 2026

Deterministic real-time simulation, native and WebAssembly

**1.6 ms** tick p50 · 64 players, 718 units

**0.8 KB/s** bandwidth per player · 0 dropped

Rust · WebAssembly · tokio · WebGL2 · Cloudflare Workers — Personal project · private repository

**Shrimp Farm Platform** · CTO & lead engineer 2024 – present

Multi-tenant AgriTech SaaS for commercial aquaculture

**Dozens** farms managed per operator

**1000s** ponds under monitoring

Next.js · Supabase · PostgreSQL · TFLite · Gemini — farms.shrimpfarm.vn

**Kèo Thơm** · Architect & full-stack engineer 2025

Real-time auction marketplace with escrowed wallets

Next.js · Supabase · PostgreSQL · Cloudflare Pages · TypeScript — keothom.bid

## CAPABILITIES

| | |
|---|---|
| **Architecture** | Distributed systems · Microservices & event-driven · Real-time & multiplayer · Multi-tenant SaaS · C4 · ADRs · data contracts · Security & threat modelling |
| **AI systems** | LLM integration (OpenAI, Gemini, Claude) · OCR & document pipelines · Agents & MCP servers · RAG & vector search · Human-in-the-loop approval · Evaluation & benchmarks |
| **Cloud & platform** | Google Cloud · AWS · Kubernetes (HPA/VPA) · Cloudflare Workers · CI/CD & IaC · Observability · OpenTelemetry |
| **Backend & data** | Node.js · NestJS · Fastify · Rust · Go · C/C++ · .NET · PostgreSQL · MySQL · Redis · Kafka · BullMQ · gRPC · REST · WebSocket · Ledgers, idempotency, outbox |
| **Leadership & clients** | Pre-sales & solution workshops · Exec & client presentations · Teams up to ~300 engineers · Delivery across Product · QA · DevOps · Code review & standards · Mentoring |

## RECOGNITION

- Winner, CodeProject Visual Studio Extension Contest
- Products reaching 100,000+ users and downloads
- First Vietnamese to solve 300+ Project Euler problems
- Reverse-engineered a 3D online game client at assembly level

Full case studies, architecture notes and benchmarks at jackle.dev